

Determining the Bulk Density and Constraining the Internal Structure of TOI-163b Through Combined Transit Photometry and Radial Velocity Analysis

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Abstract

This proposal aims to determine the bulk density and constrain the internal structure of the hot Jupiter TOI-163b using a combination of transit photometry and radial velocity analysis. Transit light curves from the TESS mission will be analyzed to derive the planetary radius and orbital period, while archival FEROS radial velocity measurements will be modeled to estimate the planetary mass. Combining these methods enables a direct determination of the planet's bulk density, providing insight into its internal composition and inflation mechanisms. We hypothesize that the independently derived mass and radius of TOI-163b will confirm that the planet belongs to the inflated hot Jupiter population and that the orbital periods derived from transit photometry and radial velocity measurements will agree within statistical uncertainties. The independently derived orbital periods from both methods will also be compared to assess the consistency and robustness of the analysis. Results will be compared with published literature values to evaluate the accuracy of the derived parameters.

Scientific Justification

As of 2026, more than 6000 exoplanets have been discovered [NASA Exoplanet Archive,], the majority of which were detected using either the transit method or the radial velocity (RV) technique. While exoplanet detection has become increasingly common, obtaining detailed constraints on planetary structure and composition remains a major challenge in modern astronomy.

The aim of this research is to combine these two complementary observational techniques to determine the bulk density of the exoplanet TOI-163b and constrain its internal structure. The radial velocity method provides an estimate of the planetary mass through measurements of Doppler shifts in the stellar spectrum, while transit photometry determines the planetary radius by measuring the fractional decrease in stellar flux during transit events. By combining these independently measured quantities, the bulk density of the planet can be derived, which serves as one of the strongest observational constraints on planetary composition.

We will investigate the exoplanet TOI-163b, a hot Jupiter orbiting the bright, metal-rich F-type star TOI-163. The system was characterized by [Kossakowski et al., 2019], who reported a planetary radius of approximately 1.49 Jupiter radii and a mass of roughly 1.22 Jupiter masses. These properties place TOI-163b within the inflated hot Jupiter population. Inflated hot Jupiters are particularly interesting because their large radii cannot be fully explained by standard planetary structure models, suggesting the presence of additional heating or inflation mechanisms.

The high stellar metallicity and close orbital separation of TOI-163b make this system especially valuable for studying planetary migration, atmospheric evolution, and star-planet interactions in highly irradiated environments. Understanding inflated hot Jupiters contributes to broader questions regarding planetary inflation mechanisms, orbital migration, atmospheric escape, and the formation of giant planets around metal-rich stars.

Transit photometry data for TOI-163b is publicly available through the Mikulski Archive for Space Telescopes [MAST,], while archival radial velocity measurements were obtained using the FEROS [(ESO),] spectrograph at the La Silla Observatory in Chile. The availability of high-quality datasets from both methods makes TOI-163b an excellent candidate for a combined observational analysis.

We hypothesize that the independently derived orbital periods from transit photometry and radial velocity measurements will agree within statistical uncertainties and that the calculated bulk density will confirm TOI-163b as an inflated gas giant with a low mean density. Agreement between both observational methods would additionally validate the robustness of the derived planetary parameters.

Technical Justification

To determine the density and internal structure of TOI-163b, we will analyze archival datasets from two high-precision observational sources: transit photometry from the TESS mission and radial velocity measurements from the FEROS spectrograph at La Silla Observatory.

The TESS photometer operates in the wavelength range of approximately 600–1000 nm and provides high-precision light curves suitable for exoplanet transit analysis. We will use the publicly available two-minute cadence observations because their high temporal resolution allows accurate characterization of transit ingress and egress features. This temporal resolution is essential for determining the transit depth, orbital period, and planetary radius with sufficient precision.

The transit light curves will be downloaded from the MAST archive and processed using the Python exoplanet package developed by [Foreman-Mackey et al., 2020]. The light curves will first be normalized and corrected for instrumental trends and outliers before transit fitting is performed. From these fits, we aim to derive the orbital period, transit depth, planetary radius ratio, and orbital inclination.

Radial velocity measurements for TOI-163b were obtained using FEROS, a high-resolution echelle spectrograph with spectral resolution $R \approx 48,000$. The FEROS spectra were reduced using the CERES pipeline described by [Brahm et al., 2017]. The expected RV semi-amplitude of TOI-163b is significantly larger than the instrumental uncertainties of approximately 20–50 m/s, ensuring that the planetary signal can be robustly detected and modeled.

We will fit a Keplerian orbital model to the radial velocity measurements under the assumption of a single planet in a circular orbit. Markov Chain Monte Carlo (MCMC) techniques will be used to estimate the posterior distributions and uncertainties of the fitted parameters, including the orbital period, RV semi-amplitude, and planetary mass.

The available FEROS dataset consists of 20 radial velocity measurements collected over approximately 47 days. These observations are sufficient to sample multiple orbital cycles, as the expected orbital period is approximately 4.23 days. The radial velocities contain a systematic velocity offset caused by instrumental calibration and the motion of the TOI-163 system itself, which will be included as a free parameter during the fitting process.

Weather and atmospheric conditions during the FEROS observations were stable, with moderate seeing and low humidity, minimizing systematic uncertainties in the radial velocity measurements. The FEROS enclosure temperature was actively controlled to reduce thermal expansion effects that could otherwise influence wavelength calibration accuracy.

$$\rho = \frac{3M}{4\pi R^3} \quad (1)$$

The resulting density will be compared with theoretical mass-radius relationships for gas giants in order to constrain the internal structure and composition of the planet. We expect the derived parameters to be consistent with previously published values of approximately:

- Orbital period: 4.231 ± 0.001 days

- Planetary radius: $1.49 \pm 0.05 R_{\text{Jup}}$
- Planetary mass: $1.22 \pm 0.15 M_{\text{Jup}}$

Although our uncertainties may be larger than those reported in previous literature due to the scope of this project, the combined analysis should still provide meaningful constraints on the planetary structure.

Potential limitations of this study include stellar activity, instrumental systematics, and the relatively small number of available radial velocity measurements. These effects may increase uncertainty in the derived planetary mass and orbital parameters.

Methodology

The analysis will be divided into four major stages.

First, TESS light curves will be downloaded from the MAST archive and preprocessed by removing outliers, correcting instrumental trends, and normalizing the stellar flux. Transit models will then be fitted to the light curves using the Python exoplanet package.

Second, archival FEROS radial velocity measurements will be modeled using a Keplerian orbital solution. An MCMC analysis will be performed to estimate parameter uncertainties and determine the best-fit orbital parameters.

Third, the independently derived orbital periods from the transit and radial velocity analyses will be compared to assess consistency between both methods.

Finally, the derived planetary mass and radius will be combined to calculate the bulk density of TOI-163b. This density will then be compared with theoretical models of gas giant planets to constrain the internal composition and inflation state of the planet.

Expected Results and Significance

This project is expected to produce an independent estimate of the mass, radius, orbital period, and bulk density of TOI-163b using publicly available observational data. The derived density will allow us to determine whether the planet is consistent with a gas-rich inflated hot Jupiter structure.

In addition to characterizing TOI-163b itself, this work demonstrates the scientific value of combining transit photometry and radial velocity measurements for exoplanet characterization. The comparison of orbital periods derived through both methods will also provide a consistency check on the robustness of the analysis.

More broadly, this project contributes to ongoing efforts to understand the formation, migration, and inflation mechanisms of hot Jupiter exoplanets in highly irradiated and metal-rich stellar environments.

References

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